

How often does R^2 pick the right release model?

A simulation study of model selection and mechanism assignment in dissolution kinetics

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dissokin: an open source Python toolkit and reproducible study

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Abstract

Release kinetics from an in vitro dissolution profile are routinely assigned by fitting a panel of models and reporting the one with the highest R^2 , and the transport mechanism is then read off the Korsmeyer-Peppas exponent n as a point estimate with no interval. This report tests both practices against known ground truth. 44 800 profiles were simulated from seven release laws across four sampling schedules and four assay noise levels, and all eight candidate models were fitted to each. Selecting by R^2 recovered the generating model in 37% of profiles; selecting by AICc recovered it in 77%. The two rules disagreed on 67% of profiles. The failure is systematic rather than noisy: where the truth had one free parameter, R^2 recovered it in 17.7% of 32 000 profiles, because a richer model can only fit at least as well and R^2 charges nothing for the extra parameter. On data generated by pure Higuchi diffusion, R^2 named Korsmeyer-Peppas in 94% of cases, which is the single most common conclusion in the applied literature. A second experiment estimated n under the conventional 60 % truncation rule, which left 4.6 usable points on a six point schedule. Whether a mechanism can be assigned turned almost entirely on distance from a Ritger-Peppas boundary: for true exponents at least 0.10 from one the classification was correct in 97% of profiles, and for exponents within 0.06 of one in 61%, which the point estimate gives no way to tell apart. The log-log linearisation used in most published analyses carried 1.8 times the root mean square error of nonlinear least squares, and the two conventions assigned different mechanisms to the same profile in 15% of cases. The f_2 similarity factor, by contrast, proved well behaved: away from the threshold its decision was correct in 99.4% of studies. A calibration experiment turned the same scepticism on the method used here and found the residual bootstrap itself undercovers at these sample sizes, 80.5% against a nominal 95 %, with rescaling of the residuals by $\sqrt{n/(n-k)}$ recovering 88.3%. The accompanying package implements information-criterion ranking with Akaike weights and bootstrap intervals as defaults.

1 The practice being tested

An in vitro release profile is a small dataset. A dissolution monograph typically specifies six to twelve sampling points, and the assay behind each one carries a standard deviation of a few percent released. Against that dataset it is standard to fit the classical release models [2, 7, 11], tabulate R^2 for each, declare the largest the “best fit model”, and, if Korsmeyer-Peppas is the winner, quote its exponent n and assign a transport mechanism from the Ritger-Peppas boundaries [8, 9]. The most widely used tool for this, DDSolver [12], reports R^2 , adjusted R^2 , AIC and the model selection criterion side by side, so the choice of rule is left to the analyst, and R^2 is the one that reaches the paper.

Two things are wrong with this in principle. First, R^2 never decreases when a parameter is added, so comparing a one-parameter model against a two-parameter model on R^2 is not a comparison at all; the richer model wins whenever the models are nested and is heavily favoured when they are not. Second, n is an estimate from a handful of points and carries sampling error, but the mechanistic boundaries it is compared against (0.45 and 0.89 for a slab) are treated as though it did not.

Neither objection is new; both are ordinary statistics. What is not established is how much they matter at the sample sizes and noise levels dissolution work actually operates at. A textbook can say that R^2 favours the richer model without telling you whether that changes the conclusion once in fifty profiles or once in three. Answering it needs the true generating model, which no real dataset provides, so the only route is simulation. That is what the rest of this report does.

2 Methods

2.1 Models

Eight release laws are implemented, each mapping time t in hours to cumulative percent released Q : zero order ($Q = k_0 t$), first order ($Q = 100[1 - e^{-k_1 t}]$), Higuchi [1, 2] ($Q = k_H \sqrt{t}$), Korsmeyer-Peppas [7] ($Q = k_{KP} t^n$), Hixson-Crowell [3] ($Q = 100[1 - (1 - k_{HC} t)^3]$), Weibull [4] ($Q = 100[1 - e^{-t^\beta/\alpha}]$), Hopfenberg [6] ($Q = 100[1 - (1 - k_{HB} t)^n]$) and Baker-Lonsdale [5], which is implicit,

$$\frac{3}{2} \left[1 - \left(1 - \frac{Q}{100} \right)^{2/3} \right] - \frac{Q}{100} = k_{BL} t, \quad (1)$$

and is inverted here by a vectorised bisection on Q rather than by the usual linearisation, so that the fit is performed in the same units as the data.

Hopfenberg reduces exactly to Hixson-Crowell at $n = 3$. In the original formulation n is fixed by device geometry and is not estimated; fitting it freely, as is common, makes the model degenerate, and on data that Hixson-Crowell describes well the optimiser drives n to the bound and the two return identical residual sums of squares. Both are retained here because both appear in the literature. Ties in R^2 are broken in favour of the model listed first, which is the simpler one. That convention is generous to R^2 , so it makes the conclusions below conservative rather than the reverse.

2.2 Selection criteria

For least squares with unknown constant variance, the maximised log likelihood gives AIC [17] and its small sample correction [19] as

$$\text{AIC} = n \ln \left(\frac{\text{RSS}}{n} \right) + 2(k+1), \quad \text{AICc} = \text{AIC} + \frac{2(k+1)(k+2)}{n-k-2}, \quad (2)$$

where k counts the model parameters and the $+1$ counts the variance. The small-sample correction is not optional here: at $n = 6$ and $k = 2$ the correction term is larger than the penalty it corrects. BIC [18] is reported alongside for comparison. Akaike weights $w_i \propto \exp(-\frac{1}{2} \Delta_i)$ [20] accompany the ranking, because a profile that cannot distinguish two models should say so rather than name a winner.

2.3 Simulation

Profiles were generated from a known model and parameters, evaluated on a sampling schedule, and perturbed by additive Gaussian noise of standard deviation σ in percent released, then clipped to $[0, 100]$ because an assay cannot report a negative or above-label release. Parameters were chosen so that each scenario reaches 75 to 96 % released over a twelve hour window, which is the regime in which the models are genuinely hard to separate. Schedules were geometric in time, matching the dense-early pattern of a real monograph, at 6, 8, 10 and 16 points; $\sigma \in \{1, 2, 3, 5\}$. Every draw is seeded, so all numbers in this report regenerate exactly.

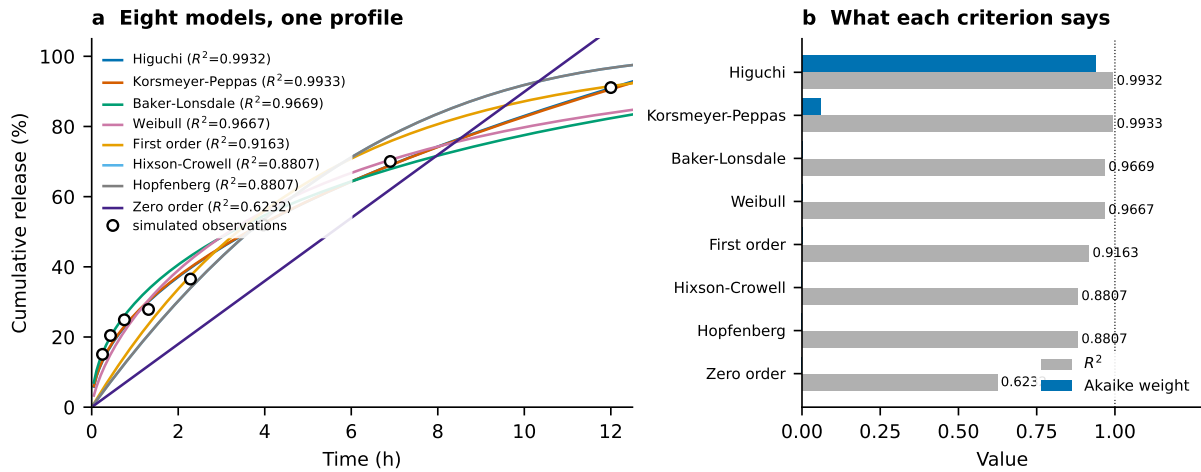


Figure 1: Eight release models are nearly indistinguishable on one profile. Data simulated from Higuchi diffusion, eight points, $\sigma = 2\%$. (a) All eight fitted curves. (b) R^2 separates the leading models in the fourth decimal; the Akaike weight does not.

2.4 The 60 percent rule

The Korsmeyer-Peppas power law is derived for the early stage of release and is conventionally fitted only below about 60 % released [8, 10]. Experiment 2 applies the rule honestly, keeping the leading run of points below the cutoff. This is not a technicality: it is what leaves a two-parameter model with a median of 6.7 points, and in the sparsest condition 4.0.

3 Experiment 1: which selection rule recovers the truth?

3.1 One profile, eight answers

Figure 1 makes the problem concrete on a single profile simulated from Higuchi diffusion at $\sigma = 2$. Korsmeyer-Peppas attains $R^2 = 0.99327$ against Higuchi's 0.99320. On the usual rule Higuchi loses, and the report would state that release follows Korsmeyer-Peppas with $n = 0.49$, indicating anomalous transport. In fact $n = 0.49$ is Higuchi ($n = \frac{1}{2}$) with an extra parameter spent, the margin is 7×10^{-5} in R^2 , and AICc places 94 % of its weight on Higuchi.

3.2 Recovery rates

Across 112 conditions and 400 replicates each (44 800 profiles), the recovery rates were R^2 37%, adjusted R^2 68%, BIC 73% and AICc 77% (Figure 2). The rules disagreed with each other on 67% of individual profiles, so this is not a distinction without a difference.

Three observations matter more than the headline numbers.

The mean best R^2 was 0.9893. A reported R^2 above 0.98 is therefore not evidence of anything; it is what every model in the panel produces on every profile in the study.

The decision is made in the fourth decimal. The median gap between the best R^2 in the panel and the R^2 of the true generating model was 0.00072, and the 90th percentile 0.0046. Papers routinely report R^2 to three decimals, which is not enough resolution to reproduce their own selection.

The failure is structural, not noisy. Where the generating model had one free parameter, R^2 recovered it in 17.7% of 32 000 profiles, against 89% for AICc. When the truth had two, R^2 recovered it 84% of the time. R^2 is not detecting the model; it is counting parameters. The mean parameter count of the R^2 -selected model was 1.86 against 1.19 for AICc (Figure 3b).

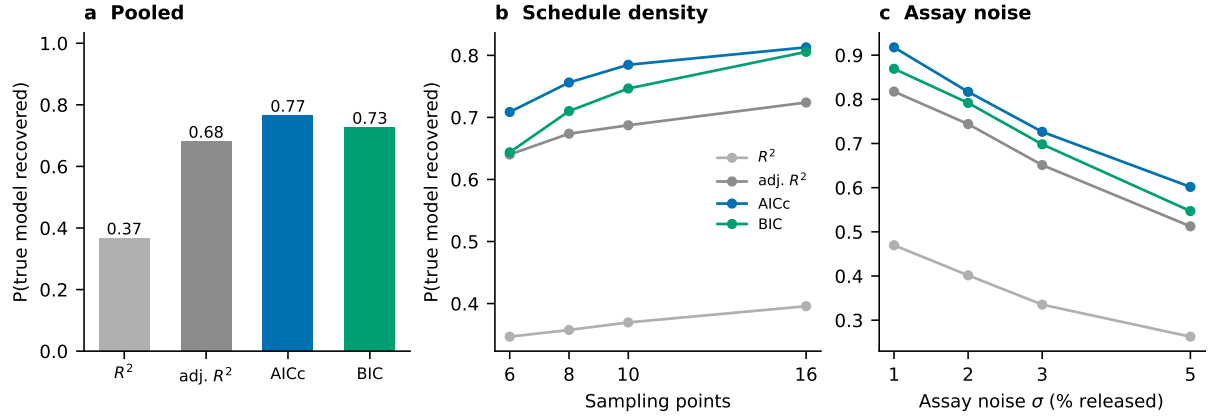


Figure 2: Recovery of the generating model by selection rule. (a) Pooled over all conditions. (b, c) Broken down by schedule density and assay noise. Denser sampling and lower noise help every rule, but do not close the gap between R^2 and the information criteria.

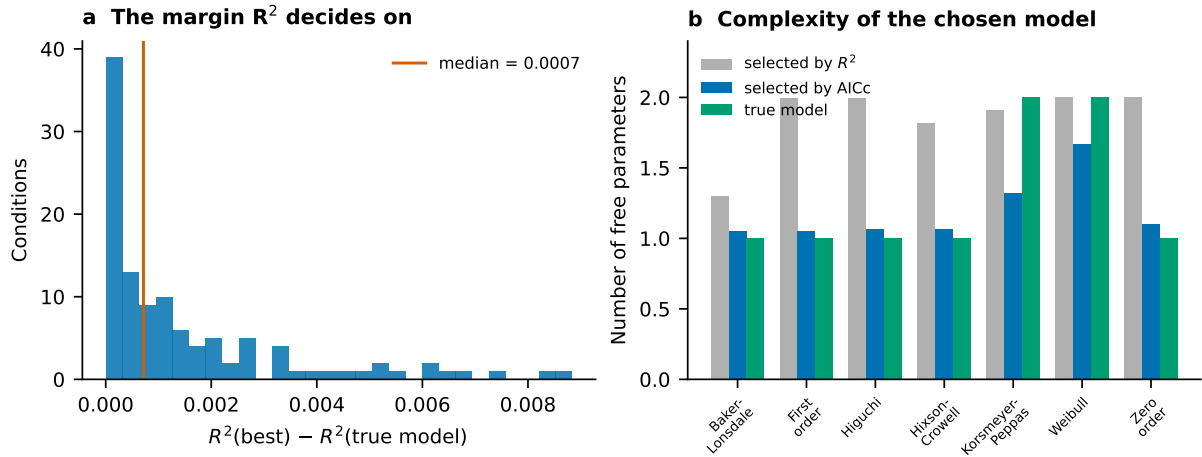


Figure 3: What R^2 is actually deciding on. (a) Distribution across conditions of the R^2 margin between the panel best and the true model. (b) Mean number of free parameters in the selected model, against the truth.

3.3 Where each rule fails

Table 1 breaks recovery down by generating model, and the pattern of errors is more informative than the rates. On Higuchi data R^2 named Korsmeyer-Peppas in 94% of profiles; on first order data it named Weibull in 96%. These are not random errors, they are the two most frequently reported conclusions in the applied dissolution literature, and this study generates both of them from data containing neither.

AICc is not uniformly better, and the case where it is worse is worth stating plainly. On Korsmeyer-Peppas data with $n = 0.42$, AICc recovered the truth only 31% of the time, naming Baker-Lonsdale instead in 62%; R^2 scored 80% here. A one-parameter model approximates a power law with n near the diffusion exponent closely enough that parsimony correctly refuses to pay for the second parameter. That is the intended behaviour of an information criterion rather than a defect, but it means AICc should be read as identifying the simplest model the data support, not as identifying physical truth.

Conditions helped all rules in the same direction and by similar amounts: AICc recovery rose from 71% at six points to 81% at sixteen, and from 60% at $\sigma = 5$ to 92% at $\sigma = 1$. No achievable sampling density rescues R^2 .

Table 1: Percentage of profiles in which each rule recovered the generating model, by truth. k is the number of free parameters in the true model. Pooled over schedules and noise levels, 400 replicates per condition.

Generating model	k	R^2	adj. R^2	AICc	BIC
Baker-Lonsdale	1	70.2	80.3	94.5	84.7
First order	1	0.0	65.7	86.7	74.4
Higuchi	1	0.0	64.7	89.2	74.5
Hixson-Crowell	1	18.3	63.6	86.0	71.5
Korsmeyer-Peppas	2	80.2	65.1	30.8	56.6
Weibull	2	88.5	82.3	61.9	79.5
Zero order	1	0.0	55.2	86.9	67.3

4 Experiment 2: can a mechanism be read off n ?

4.1 Design

Profiles were generated from the power law with a known exponent $n \in \{0.35, 0.45, 0.60, 0.75, 0.95\}$, spanning the quasi-Fickian, anomalous and super case II regions and sitting on the boundaries themselves. Each replicate was truncated at 60 % released, and n was then estimated both by nonlinear least squares on Q and by the log-log linear regression used in most published analyses. A residual bootstrap [21, 22] of 200 resamples gave an interval for the nonlinear estimate. 60 conditions, 150 replicates each.

4.2 What limits a mechanistic assignment

The truncation rule bites harder than the schedule suggests, and this was the first thing that surprised me. Twelve scheduled points leave 9.4 below the cutoff on average and six leave 4.6, so a schedule that looks twice as dense yields roughly twice as many usable points but all of them crowded into the early part of the curve, where the power law is flattest and n is least well determined.

The mean 95 % bootstrap interval for n was 0.17 wide, against an anomalous band of 0.44 (Figure 4b). That is the first surprise: on average the interval is comfortably narrower than the class it has to fit inside, so the blanket claim that n is never precise enough to support a mechanism is not what the data say. Nevertheless the interval lay entirely inside one mechanistic class in only 47% of profiles.

The two facts are reconciled by where the true exponent sits (Figure 4c). Splitting the conditions by distance from a Ritger-Peppas boundary separates them almost completely (Table 2). For true exponents at least 0.10 from a boundary ($n = 0.35, 0.60, 0.75$) the point estimate landed in the correct class 97% of the time and the interval fitted inside one class in 68%. For exponents on or within 0.06 of a boundary ($n = 0.45, 0.95$) the same numbers fall to 61% and 17%. At $n = 0.45$, which is the slab boundary itself, the classification is a coin toss, as it must be.

This is a more useful conclusion than a blanket one, and it is not the one I expected. A mechanism read off n is reliable when n falls well inside a class, and close to worthless when it falls near a boundary. The catch is that the analyst cannot tell which situation applies from the point estimate alone, because a value of 0.47 looks exactly as authoritative as a value of 0.60. Computing the interval is what separates them, and it costs seconds.

4.3 The intervals are narrower than they should be

Bootstrap coverage of the true n was 79.8% against a nominal 95 %. The shortfall tracked the number of points (73% on a six point schedule, 80% on eight, 87% on twelve) and was essentially flat in σ . That pattern points at degrees of freedom rather than at noise. Fitted residuals are shrunk relative to the true errors because the fit has already absorbed k of the n degrees of freedom, and with four to six usable points and two parameters that is a large fraction of them.

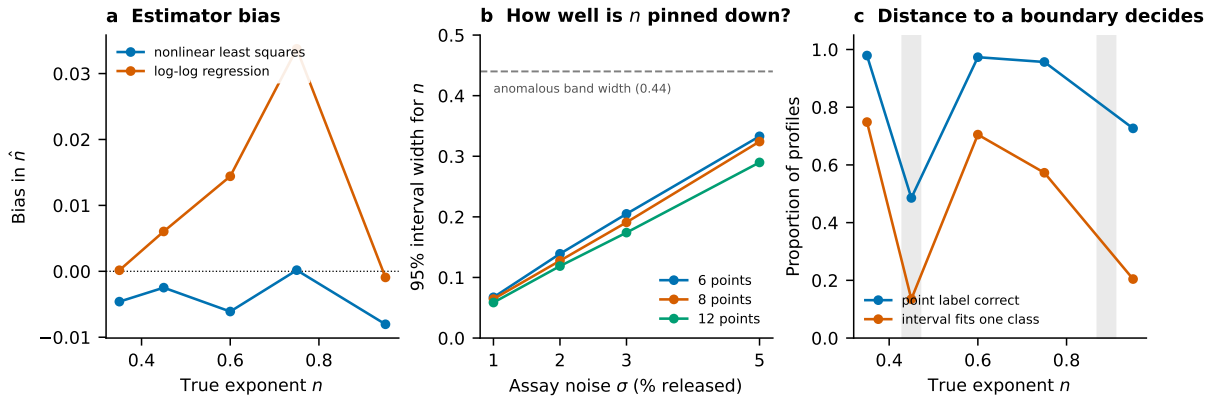


Figure 4: Estimating the Peppas exponent under the 60 % rule. (a) Bias of the two estimators against the true exponent. (b) Width of the 95 % bootstrap interval for n ; the dashed line is the width of the whole anomalous transport band, so on average the interval is narrower than the class it must fit inside. (c) Reliability against the true exponent, with the Ritger-Peppas boundaries shaded: what decides whether a mechanism can be assigned is not precision in general but distance from a boundary.

A dedicated calibration experiment confirms the diagnosis. At $n = 0.60$ and $\sigma = 2$, with 200 replicates and 250 resamples, the plain residual bootstrap covered the truth 80.5% of the time and produced intervals 0.76 times the width of the estimator’s actual sampling distribution. Rescaling the residuals by $\sqrt{n/(n-k)}$ before resampling raised coverage to 88.3% at a width ratio of 0.96, with the largest gain on the sparsest schedule (71.5% to 87.0% at six points). The package applies the rescaling by default; Experiment 2 was run without it, so the interval widths and single-class proportions reported above are optimistic, and the true rate at which a profile supports a single mechanistic label is lower than 47%.

I could have switched the correction on before running anything and reported only the corrected numbers. Reporting both seemed more honest, and more useful: an interval that undercovers is worse than no interval at all, because it still carries the authority of a number.

4.4 The linearisation is not free

Taking logs of both sides of $Q = k_{KP}t^n$ and regressing $\ln Q$ on $\ln t$ is convenient and nearly universal. It is not equivalent to fitting the model to the data, because the transform reweights the points: if the assay error is roughly constant in percent released, the early low points, where the relative error is largest, are given the most leverage. Averaged over all conditions the bias was modest for both, +0.011 for the linearisation against -0.004 for nonlinear least squares; the cost shows up in the spread rather than the centre. Root mean square error was 0.116 against 0.063, so the linearisation carries 1.8 times the error of fitting the model to the data it was measured in. The two conventions assigned different mechanistic labels to the same profile in 15% of cases, so on roughly one profile in 7 the reported mechanism is determined by the fitting convention rather than by the data.

4.5 A worked profile

Figure 5 takes one simulated profile through the whole pipeline: eight scheduled points, of which the 60 % rule keeps a subset, a nonlinear fit, and 3000 bootstrap resamples of n plotted against the mechanistic boundaries. The output is not a mechanism. It is an interval, and the honest reporting of it is the interval together with the proportion of the resampling distribution falling in each class.

Table 2: Estimation of the Peppas exponent by true value, pooled over schedules and noise. NLS: nonlinear least squares on percent released; LL: log-log linear regression. The last three columns are percentages of profiles. 150 replicates and 200 bootstrap resamples per condition.

n	True class	Bias in $\hat{n}$		95% CI width	CI in one class (%)	Correct class (%)	
		NLS	LL			NLS	LL
0.35	Quasi-Fickian diffusion	-0.005	+0.000	0.11	75	98	96
0.45	Anomalous	-0.002	+0.006	0.13	14	49	51
0.60	Anomalous	-0.006	+0.014	0.16	70	97	93
0.75	Anomalous	+0.000	+0.034	0.20	57	96	84
0.95	Super Case II transport	-0.008	-0.001	0.27	20	73	60

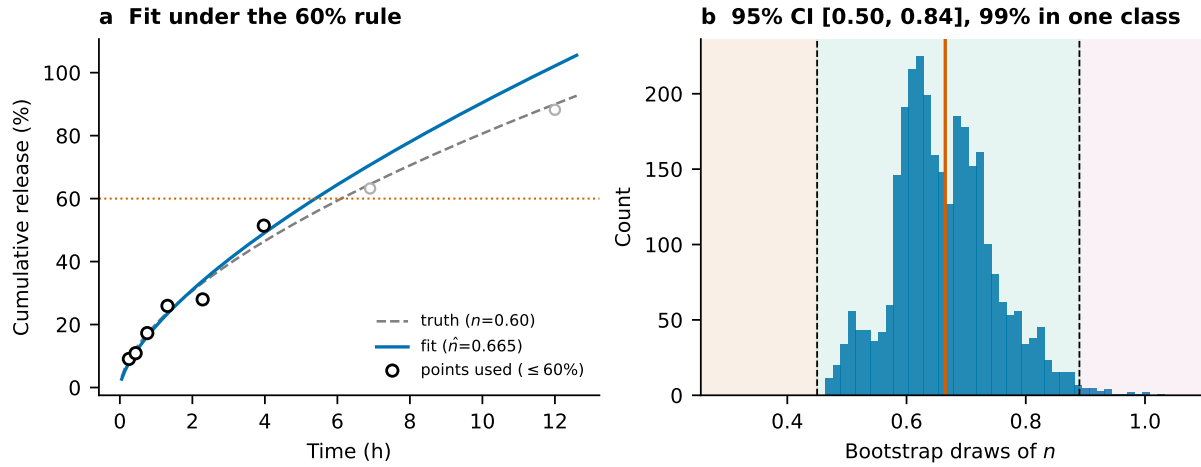


Figure 5: One profile reported properly. (a) The fit, with points excluded by the 60 % rule shown in grey. (b) The bootstrap distribution of n against the Ritger-Peppas boundaries, shaded by mechanistic class.

5 Experiment 3: is the f_2 decision stable?

The same scepticism was applied to the similarity factor $f_2 = 50 \log_{10}(100/\sqrt{1 + (R - T)^2})$ [13], which is computed from the means of a small number of vessels and is compared against a fixed threshold of 50 by both the FDA and the EMA [15, 16]. Its sampling behaviour has been examined before [14]; the question here is only whether the decision it produces is stable at the vessel counts actually used. It survives the test, which is worth saying plainly given how the rest of the report has gone. Across 2 000 simulated studies per condition, the pass or fail decision matched the truth in 99.4% of studies whenever the true f_2 lay more than one unit from the threshold, with a worst case of 90.9%; the standard deviation of the reported f_2 at twelve vessels was under 0.59 units in the region that matters. Within one unit of the threshold the decision is a coin flip, which is unavoidable for any sharp cutoff and is not a defect of the statistic.

One real effect is worth reporting. On genuinely identical profiles f_2 is biased downward, because assay noise inflates the mean squared difference: mean f_2 fell to 75 rather than 100 at 6 % vessel standard deviation with 6 vessels. The bias is conservative, in that it pushes toward declaring difference, and it is negligible once the profiles genuinely differ (+0.11 on average across all non-identical cases). A bootstrap interval remains worth reporting near the threshold: in the worked example the point estimate was $f_2 = 50.2$, nominally a pass, with a 95 % interval of [47.7, 52.4] and 49% of resamples below 50.

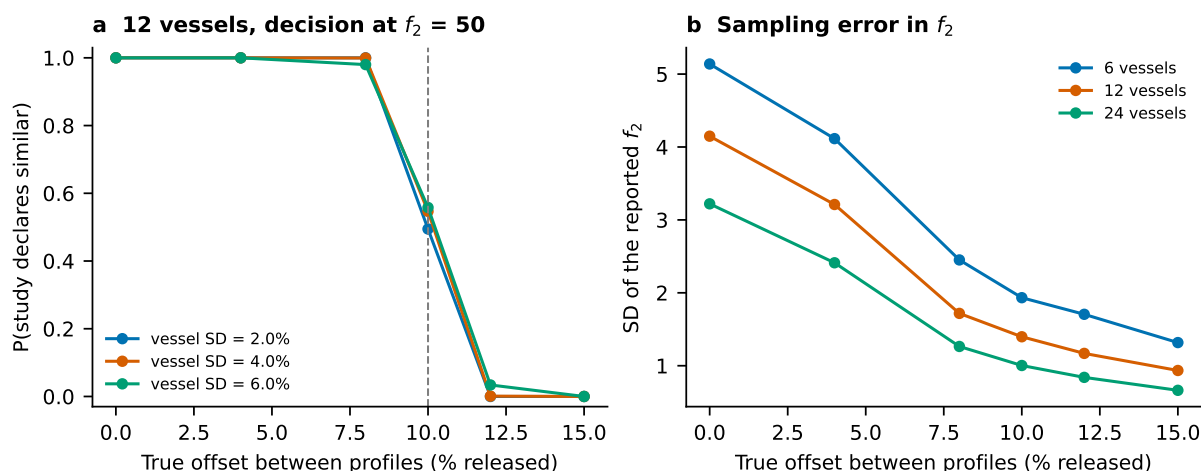


Figure 6: f_2 is well behaved away from its threshold. (a) Probability that a twelve-vessel study declares similarity, against the true offset between formulations. (b) Sampling standard deviation of the reported f_2 .

6 What to report instead

Nothing that follows is expensive, and none of it requires abandoning a model or a workflow that already exists.

Rank by AICc, not R^2 , and report Akaike weights. The computation is three lines on top of a fit that has already been performed. Where two models carry comparable weight, that is the result, and naming one of them a winner discards it.

Report R^2 if a journal requires it, but not as the selection criterion, and quote it to at least four decimals if it is quoted at all, since the selection turns on the fourth.

Give n an interval, and rescale the residuals. A residual bootstrap of a few thousand resamples takes seconds, and the $\sqrt{n/(n-k)}$ correction is one line. If the interval spans a Ritger-Peppas boundary, the correct statement is that the data do not distinguish the mechanisms, and that statement is more useful than a label the data cannot support. If the interval sits well inside one class, the label is supported and can be stated with confidence. That is the other half of the result, and the reason to compute the interval rather than to assume the worst.

Fit the power law to Q , not to $\ln Q$, or state that the linearisation was used and accept the bias it introduces.

Say how many points survived the 60 % truncation. Two parameters fitted to four points is a different claim from two parameters fitted to ten, and the distinction disappears from a paper that only reports the full schedule.

7 Limitations

The study is simulation only, and deliberately so: recovery of a generating model cannot be measured where the truth is unknown, which rules out real profiles for this question. The consequence is that the noise model is an assumption. Additive Gaussian error of constant variance is a reasonable description of a UV assay over the working range, but real dissolution error is heteroscedastic, larger at the low end and at the sampling times where manual withdrawal dominates, and errors within a vessel are correlated across time in a way independent draws do not reproduce. Both would widen intervals further, so the results here are optimistic rather than pessimistic.

The candidate panel is fixed at eight models and contains one degenerate pair, as described above. Only a slab geometry was used for the Ritger-Peppas boundaries; a spherical device shifts them to 0.43 and 0.85 and narrows the anomalous band, which would lower the single-class proportion further. Parameter ranges

were chosen to give plausible release extents, and a panel of formulations that release only 40 % over the window would be a different and probably harder problem.

The calibration experiment was run at a single exponent and noise level, so the rescaled coverage quoted here should be read as evidence that the diagnosis is right rather than as a general guarantee; residual rescaling is a first order fix, and bias-corrected or accelerated intervals would be the next step. The undercoverage also means the single-class proportions reported above are upper bounds, which is stated where they appear but is worth repeating: mechanism assignment near a boundary is at least as unreliable as reported, not less.

8 Software

All of the above is implemented in `dissokin`, a small Python package (`numpy`, `scipy`) providing the eight models behind one interface, least squares fitting with AIC, AICc, BIC and Akaike weights, residual and case bootstrap for any parameter, mechanism classification with its resampling distribution attached, and f_1 and f_2 with a vessel-level bootstrap. The package carries a unit test suite including analytic checks: Baker-Lonsdale must invert its own implicit equation, that Akaike weights sum to one, that no model releases drug backwards. Every experiment, figure and number in this report is regenerated by the scripts in `examples/`, and every number quoted above is inserted into this document from the stored results rather than typed in.

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Scope

This is an independent methodological study and software project, carried out after the author's Master's degree and unconnected to it. It is not a peer-reviewed publication, it has not been submitted to any journal, and it reports no experimental data. Every number reported here is produced by the accompanying code rather than asserted, and the scripts that produce them are in the repository.